Hints: Limits – 9. Sequences defined by integrals and sums

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The measure $(n+1)x^n\,dx$ on $[0,1]$ has total mass 1 and concentrates at $x=1$. Write $n\int_0^1 x^n g - \frac{n}{n+1}g(1) = n\int_0^1 x^n\big(g(x)-g(1)\big)dx$.

Hint 2. Split at $1-\delta$. Near 1, $|g(x)-g(1)|<\varepsilon$ by continuity, which contributes at most ε. On $[0,1-\delta]$ the contribution is at most $2\max|g|\cdot n(1-\delta)^{n+1}\to 0$.

Hint 3. For $g\in C^1$, integrate by parts: $\int_0^1 x^n g = \frac{g(1)}{n+1} - \frac{1}{n+1}\int_0^1 x^{n+1}g'$, and the last integral is $\frac{g'(1)}{n+2}+o(\frac{1}{n})$ by the first part. Hence $n\int_0^1 x^n g = g(1) - \frac{g(1)+g'(1)}{n} + o(\frac{1}{n})$; for $g\in C^2$ the error is $O(n^{-2})$.

Problem 2

Hint 1. Both integrals can be computed exactly: substitute $y=nx$.

Hint 2. $\int_0^1 \frac{n\,dx}{1+n^2x^2} = \arctan n \to \frac{\pi}{2}$, although the integrand tends to 0 for every $x>0$. The mass $\frac{\pi}{2}$ escapes into a shrinking neighbourhood of 0, and no integrable function dominates the sequence ($\sup_n \frac{n}{1+n^2x^2} = \frac{1}{2x}$).

Hint 3. $\int_0^1 \frac{nx\,dx}{1+n^2x^2} = \frac{\ln(1+n^2)}{2n}\to 0$. Here the integrands are bounded by $\frac{1}{2}$ and tend to 0 pointwise, so dominated convergence applies, and the limit and the integral commute.

Problem 3

Hint 1. This is Theorem 2 of the class work with the 1-periodic function $g(y)=\{y\}$, whose mean is $\frac{1}{2}$. A direct proof is just as short.

Hint 2. Split $[0,1]$ into the intervals $[\frac{k-1}{n},\frac{k}{n}]$. On each, $\int\{nx\}\,dx = \frac{1}{2n}$, and f differs from $f(\frac{k}{n})$ by at most $\omega(\frac{1}{n})$, the modulus of continuity.

Hint 3. So $\int_0^1 f(x)\{nx\}dx = \frac{1}{2}\cdot\frac{1}{n}\sum_k f(\frac{k}{n}) + O(\omega(\frac{1}{n}))$, which tends to $\frac{1}{2}\int_0^1 f$ by uniform continuity and Riemann sums.

Problem 4

Hint 1. Let $M=\max f$. The upper bound $(\int f^n)^{1/n}\leqslant M$ is trivial. For the lower bound, take an interval of length $\delta>0$ on which $f\geqslant M-\varepsilon$.

Hint 2. Then $(\int_0^1 f^n)^{1/n}\geqslant \delta^{1/n}(M-\varepsilon)\to M-\varepsilon$.

Hint 3. Refinement (Laplace's method). Let $f\in C^2$ with a unique maximum at an interior point x_0 and $f''(x_0)<0$. Write $f^n = M^n e^{n(\ln f - \ln M)}$ with $\ln f(x)-\ln M = -\frac{|f''(x_0)|}{2M}(x-x_0)^2(1+o(1))$. As in Problem 5 of the class work, split at $|x-x_0|=n^{-1/3}$; outside, $f\leqslant M-c$.

Hint 4. The answer is $\int_0^1 f^n \sim f(x_0)^n\sqrt{\frac{2\pi f(x_0)}{n|f''(x_0)|}}$.

Problem 5

Hint 1. The weight x^n concentrates the integral near $x = \pi$, where $\sin x$ vanishes linearly. Substitute $x = \pi - t$: $a_n = \int_0^\pi (\pi - t)^n \sin t \, dt$.

Hint 2. $(\pi - t)^n = \pi^n e^{n \ln(1 - t/\pi)}$, which is close to $\pi^n e^{-nt/\pi}$ for small t, and $\sin t \approx t$. After $t = \pi s/n$ you get $a_n \sim \pi^n \int_0^\infty t \, e^{-nt/\pi} dt = \frac{\pi^{n+2}}{n^2}$. (Justify the approximation by splitting at a small t; or use dominated convergence after the substitution, with $(1 - \frac{s}{n})^n \leqslant e^{-s}$.)

Hint 3. Then $\frac{a_{n+1}}{a_n} \to \pi$, so $\frac{a_{n+1}}{n a_n} \sim \frac{\pi}{n} \to 0$.

Problem 6

Note. Only the values of f on $[0, 1]$ enter the sum, so the hypotheses about $[0, \infty)$ and the convergence of $\int_0^\infty f$ play no role; the statement is simply the convergence of Riemann sums for a continuous f on $[0, 1]$.

Hint 1. A continuous f on $[0, 1]$ is uniformly continuous. On each $[\frac{k-1}{n}, \frac{k}{n}]$ compare $\frac{1}{n} f(\frac{k}{n})$ with $\int f$: the difference is at most $\frac{1}{n}\omega(\frac{1}{n})$.

Hint 2. Summing gives total error $\leqslant \omega(\frac{1}{n}) \to 0$.

Hint 3. For the counterexample, take a function that is Lebesgue integrable but whose values at rational points are untypical. The indicator of the irrationals in $[0, 1]$ has $\int_0^1 f = 1$ (Lebesgue), but every $\frac{k}{n}$ is rational, so all the Riemann sums are 0.

Problem 7

Hint 1. This is the error term of Theorem 1 of the class work. Work on one interval $[\frac{k-1}{n}, \frac{k}{n}]$ and compare $\frac{1}{n} f(\frac{k}{n})$ with the integral, using Taylor's formula at the right endpoint.

Hint 2. $\int_{(k-1)/n}^{k/n} f = \frac{1}{n} f(\frac{k}{n}) - \frac{1}{2n^2} f'(\frac{k}{n}) + O(n^{-3})$, with the constant in O depending on $\max |f''|$. Summing, $n(\frac{1}{n}\sum f(\frac{k}{n}) - \int f) = \frac{1}{2n}\sum_k f'(\frac{k}{n}) + O(\frac{1}{n})$, which tends to $\frac{1}{2}\int_0^1 f' = \frac{f(1) - f(0)}{2}$.

Hint 3. For $f(x) = x^2$: $\frac{1}{n^3}\sum k^2 = \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}$, so the limit is $\frac{1}{2} = \frac{f(1) - f(0)}{2}$.

Problem 8

Hint 1. $\frac{1}{1+x^n} \to 1$ for $x \in [0, 1)$ and the integrand is bounded, so $I_n \to 1$. Then write $1 - I_n = \int_0^1 \frac{x^n}{1+x^n} dx$.

Hint 2. Substitute $t = x^n$, so $dx = \frac{1}{n} t^{1/n - 1} dt$. You get $1 - I_n = \frac{1}{n}\int_0^1 \frac{t^{1/n}}{1+t} dt$, and the integral tends to $\ln 2$. Thus $1 - I_n \sim \frac{\ln 2}{n}$.

Hint 3. For the next term, $t^{1/n} = 1 + \frac{\ln t}{n} + O(\frac{\ln^2 t}{n^2})$ and $\int_0^1 \frac{\ln t}{1+t} dt = -\frac{\pi^2}{12}$. So $1 - I_n = \frac{\ln 2}{n} - \frac{\pi^2}{12n^2} + O(n^{-3})$.

Problem 9

Hint 1. Laplace's method: $(1 - x^2)^n = e^{n \ln(1 - x^2)}$, which is close to e^{-nx^2} near 0. Alternatively, use the bounds $e^{-nx^2 - nx^4} \leqslant (1 - x^2)^n \leqslant e^{-nx^2}$ for small x, or substitute $x = \sin\theta$ to get the integral a_{2n+1} of HW 6, Problem 5.

Hint 2. After $x = s/\sqrt{n}$ the integral is $\frac{1}{\sqrt{n}}(\int_0^\infty e^{-s^2} ds + o(1)) = \frac{\sqrt{\pi}}{2\sqrt{n}}(1 + o(1))$. The same holds for a real exponent $m \to \infty$ in place of n.

Hint 3. For the ball, slice B^n by the last coordinate: $V_n = V_{n-1}\int_{-1}^{1}(1-x^2)^{(n-1)/2}dx \sim V_{n-1}\sqrt{\frac{2\pi}{n}}$. The ratio V_n/V_{n-1} tends to 0, so $V_n/2^n$ (ball over the cube $[-1,1]^n$) decays faster than any geometric sequence. In fact $V_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)} \approx \frac{1}{\sqrt{\pi n}}\big(\frac{2\pi e}{n}\big)^{n/2}$.

Problem 10

Hint 1. First prove the statement for $f(x) = e^{imx}$ with $m \in \mathbb{Z}$. For $m = 0$ it is trivial; for $m \neq 0$, $e^{im\alpha} \neq 1$ because $\frac{\alpha}{2\pi}$ is irrational, so sum the geometric progression.

Hint 2. $\big|\frac{1}{n}\sum_{k=1}^{n} e^{imk\alpha}\big| \leqslant \frac{2}{n|e^{im\alpha}-1|} \to 0 = \frac{1}{2\pi}\int_0^{2\pi} e^{imx}dx$. By linearity the statement holds for all trigonometric polynomials.

Hint 3. A continuous 2π-periodic f is a uniform limit of trigonometric polynomials (Weierstrass, or Fejér). If $\|f - P\|_\infty < \varepsilon$, both the averages and the integrals change by less than ε. This is Weyl's criterion: the points $\{\frac{k\alpha}{2\pi}\}$ are equidistributed in $[0,1)$. Approximating the indicator of an arc from above and below by continuous functions gives the equidistribution statement of the trigonometry sheet.